3.7 Equivalent Expressions

Lesson Introduction

Benchmark	MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.	Learning Target	• I can determine equivalency using the laws of exponents.

	Building On	Working Toward
Benchmark Coherence	MA.7.NSO 1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.	MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Essential Questions	• How do you determine if expressions are equivalent using the laws of exponents if the expressions do not have the same base? • How do you use the laws of exponents to rewrite numerical expressions to have a common base?
Lesson Narrative	Students explore how rewriting exponential expressions can be equivalent if they have different bases. Students use the laws of exponents to rewrite exponential expressions to determine if expressions are equivalent.

Component Summary				
	3.7.1 Warm-Up (~5 min)	Evaluate expressions with integer exponents *(SE p. 135)*	3.7.4 Guided Practice (10 min)	Rewrite expressions with different bases using laws of exponents and assess equivalence *(SE p. 137)*
	3.7.2 Exploration (5-10 min)	Explore expressions with exponents that have different bases to discover equivalence *(SE p. 135)*	3.7.5 Collaboration (10-15 min)	Rewrite expressions with as few bases as possible *(SE p. 138)*
	3.7.3 Guided Instruction (15 min)	Rewrite expressions with different bases *(SE p. 136)*	3.7.6 Wrap-Up (~5 min)	Students summarize their learning *(SE p. 139)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• Desmos scientific calculator	• Consider strategically grouping students for 3.7.5 Collaboration based on observational data from the previous lessons.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may not recognize exponents with different bases that can be rewritten with the same base.	• Students will need calculators for 3.7.2 Exploration. They do not need calculators for the following activities.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Collect and Display As students complete 3.7.5 Collaboration, collect and display student work. When displaying, organize the student work by each expression. Engage students in a discussion about what they notice in the strategies used for rewriting each of the same expressions. Look for opportunities have students recognize when students took different steps to rewrite expressions and discuss if the resulting expressions were equivalent or if they were equal. Ask students to share out if they feel they used a unique strategy.	**MA.K12.MTR.4.1: Discussion and Reflection** 3.7.3 Guided Instruction asks students to reflect and discuss how expressions with different bases were rewritten and simplified using the laws of exponents. Prompt students who used different steps to share and discuss similarities and differences between their strategies.

Differentiate Instruction

Provide students the option to verbally explain if they do or do not agree with Xavier in 3.7.4 Guided Practice.

Use observational data from the 3.7.2 Exploration, 3.7.3 Guided Instruction, and 3.7.4 Guided Practice to determine strategic grouping for the 3.7.5 Collaboration. If necessary, pull a small group of students to provide remediation or additional support during the 3.7.5 Collaboration.

Additional differentiation opportunities are denoted throughout the lesson guide.

Expression A	Expression B
$3^2 \cdot 6^3$	$\frac{(3 \cdot 2)^5}{2^2}$
$= 3^2 \cdot (\boxed{3} \cdot 2)^3$	$= \frac{3^{\boxed{5}} \cdot 2^{\boxed{5}}}{2^2}$
$= 3^2 \cdot \boxed{3}^{\boxed{3}} \cdot \boxed{2}^{\boxed{3}}$	$= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$
$= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$	

Lesson Notes:

3.7.1 Warm-Up: Bell Work

Component Overview: *Students demonstrate their conceptual understanding and mathematical fluency of evaluating numerical expressions with integer exponents.*

3.7.1 Warm-Up: Bell Work

1. Evaluate each expression. Show your work.

   a. $\frac{(-4)^2(-4)^{-3}}{-4}$

   $= \frac{(-4)^{-1}}{-4}$

   $= \frac{1}{-4 \cdot -4}$

   $= \frac{1}{16}$

   b. $\left(\left(\frac{2}{5}\right)^3\right)^{-2} \cdot \left(\frac{5}{2}\right)^{-4}$

   $= \left(\frac{2}{5}\right)^{-6} \cdot \left(\frac{5}{2}\right)^{-4}$

   $= \left(\frac{2}{5}\right)^{-6} \cdot \left(\frac{2}{5}\right)^{4}$

   $= \left(\frac{2}{5}\right)^{-2}$

   $= \left(\frac{5}{2}\right)^{2}$

   $= \frac{25}{4}$

Use student understanding to inform your instruction during this lesson. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

3.7.2 Exploration: Exponential Expressions with Different Bases

Component Overview: *Students evaluate exponential expressions with different bases and discover that they are equivalent. Provide students with calculators to evaluate the equivalent expressions.*

3.7.2 Exploration: Exponential Expressions with Different Bases

1. Three expressions are given in the table.

   a. For each given expression, generate an equivalent expression by applying exponent laws. Then, find the value of the expression using a calculator.

Given Expression	Equivalent Expression	Value
$\frac{2^{14}}{2^4}$	2^{10}	1,024
$4^7 \cdot 4^{-2}$	4^5	1,024
$16 \cdot 4 \cdot 4^2$	$16 \cdot 4 \cdot 16$	1,024

   b. Discuss your results with your partner.
   What do you notice about the three expressions?
   Answers will vary. All three expressions have the same value.

Demonstrate and explain how to enter exponential expressions in a calculator. Consider having all students use the same calculator, such as the Desmos scientific calculator.

Students may make errors during the 3.7.2 Exploration when finding an equivalent expression. For example: students may put 8 as the equivalence of 4^2. Allow the opportunity for them to correct their own thinking as they work through with their partner. Do not be afraid to let the mistakes happen; discuss them during the transition to the next component.

***Component Overview**: Students are guided through rewriting exponential expressions with different bases to have a common base.*

3.7.3 Guided Instruction: Rewriting Expressions with Different Bases

Exponent laws can be applied to exponents with the **same base.** However, sometimes a base can be rewritten to create a common base when perfect squares, cubes, fourths, etc. are present.

1. Complete the table. The first row is started for you.

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
4	$4^2 = 16$	$4^3 = 64$	$4^4 = 256$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

2. Write an exponential expression equivalent to 8^5 that has a base of 2.

Step	Description
8^5	Determine an expression with base 2 that is equivalent to 8.
$= (2^3)^5$	Substitute the power of 2 for 8 in the expression.
$= 2^{\boxed{15}}$	Apply power of a power law of exponents.

3. Write an exponential expression equivalent to 9^4 that has a base of 3.

9^4
$= (3^2)^4$
$= 3^{\boxed{8}}$

After students have completed question 1, see if students are able to rewrite 343 as an equivalent expression with an exponent.

If students are ready, ask them to rewrite 256 two different ways as an exponential expression. Responses might be 2^8 and 16^2 and 4^4.

3.7.3 Guided Instruction: Rewriting Expressions with Different Bases (continued)

4. Complete the steps to write an exponential expression equivalent to $2^3 \cdot 4^3$ that has a base of 2.

$$2^3 \cdot 4^3$$
$$= 2^3 \cdot (\boxed{2^2})^3$$
$$= 2^3 \cdot 2^{\boxed{6}}$$
$$= 2^{\boxed{9}}$$

5. Work with your partner to rewrite each expression from the Exploration using a common base of 2.

Expression A	Expression B	Expression C
$\frac{2^{14}}{2^4}$ 2^{10}	$4^7 \cdot 4^{-2}$ $(2^2)^7 \cdot (2^2)^{-2}$ $2^{14} \cdot (2^{-4})$ 2^{10}	$16 \cdot 4 \cdot 4^2$ $2^4 \cdot (2^2) \cdot (2^2)^2$ $2^4 \cdot (2^2) \cdot 2^4$ 2^{10}

Consider having students list the exponent laws used during the process to write equivalent expressions for Questions 3 and 4.

3.7.4 Guided Practice: Different Bases, Same Value

***Component Overview**: Students use the laws of exponents to rewrite exponential expressions with different bases and then determine equivalence.*

3.7.4 Guided Practice: Different Bases, Same Value

The laws of exponents can be applied to determine if expressions are equivalent.

1. Consider the expressions $3^2 \cdot 6^3$ and $\frac{(3\cdot 2)^5}{2^2}$.

   a. Complete the steps to rewrite each expression using laws of exponents.

Expression A	Expression B
$3^2 \cdot 6^3$ $= 3^2 \cdot (\boxed{3} \cdot 2)^3$ $= 3^2 \cdot \boxed{3}^{\boxed{3}} \cdot \boxed{2}^{\boxed{3}}$ $= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$	$\frac{(3\cdot 2)^5}{2^2}$ $= \frac{3^{\boxed{5}} \cdot 2^{\boxed{5}}}{2^2}$ $= 3^{\boxed{5}} \cdot 2^{\boxed{3}}$

   b. Complete the statement.

   The expressions $3^2 \cdot 6^3$ and $\frac{(3\cdot 2)^5}{2^2}$ are

   ○ equivalent
   ○ not equivalent

   because they simplify to

   ○ the same expression.
   ○ different expressions.

Allow informal observation throughout the first three activities to dictate how "guided" this 3.7.4 Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

3.7.4 Guided Practice: Different Bases, Same Value (continued)

2. Xavier says the expressions $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both expressions can be rewritten as 2^{-7}.

   a. Generate an equivalent expression for each expression using a base of 2. Show your work.

$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$	$\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$
$4^{-2} \cdot 2^{-3}$	$2^{-4} \cdot 4^{-1}$
$(2^2)^{-2} \cdot 2^{-3}$	$2^{-4} \cdot (2^2)^{-1}$
$2^{-4} \cdot 2^{-3}$	$2^{-4} \cdot 2^{-2}$
2^{-7}	2^{-6}

   b. Explain whether or not you agree with Xavier.

   I disagree with Xavier because the two expressions will not both simplify to 2^{-7}. The second expression simplifies to 2^{-6} not 2^{-7}.

If students are ready for a challenge, ask them to determine whether the following expressions are equivalent.
$\frac{6^2 \cdot 10^{-7}}{30^{-7}}$ *and* $3^7 \cdot (3 \cdot 2)^2$

3.7.5 Collaboration: Equivalent Expressions

Component Overview*: Students generate equivalent exponential expressions using as few bases as possible. Then students compare their expressions with their partner. Prompt students to discuss the law of exponents they used at each step and whether they agree with the strategy their partner used.*

3.7.5 Collaboration: Equivalent Expressions

1. With your partner, determine who is Partner A and who is Partner B.

   a. Generate an equivalent exponential expression for each expression in your column using as few bases as possible.

Partner A		Partner B
$3^3 \cdot 3^4$	3^7	$\frac{3^9}{3^2}$
$\frac{2^{-2}}{2^{-4}} \cdot 7$	$2^2 \cdot 7$	$\frac{(7 \cdot 2)^2}{7}$
$2^2 \cdot 4^2$	4^3 *or* 2^6	$4^5 \cdot 4^{-2}$
$\frac{3^5}{6^2}$	$\frac{3^3}{2^2}$	$3 \cdot \left(\frac{2}{3}\right)^{-2}$
$\frac{5^2 \cdot 5^0}{2^{-4} \cdot 2^3}$	$2 \cdot 5^2$	$\frac{10^4}{5^2 \cdot 2^3}$

   b. The expressions in each row are equivalent. If your expression and your partner's expressions are not equivalent, work together to find and correct any error(s). Then, write the equivalent expression in the middle column.

Collect and Display
Collect and display student work. Organize by each expression. Then engage students in a discussion about which laws of exponents they used and what they notice about the expressions when different laws were used.

Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the 3.7.6 Wrap-Up. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear or to address common misconceptions that were evident among the majority of the class.

Do you agree with the steps your partner took? Is there an alternative sequence of steps possible to generate the same equivalent expression? How do you know that your expression includes the fewest number of bases possible?

3.7.6 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

3.7.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

   Answers will vary.

Consider providing students with sentence stems and/or answer choices. Prompt students to look back at their work on each component of the lesson as they compose their responses.